



# Emerging integrated photonic technologies leveraging multimaterial integration for AI and datacenter applications [Invited]

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**Abstract:** Since the inception of integrated photonics, multimaterial integration has served as a primary avenue for new technology innovations. Now, with an ever-increasing demand for integrated photonics as a platform for both high-performance links from/within datacenters and AI acceleration, multimaterial integration has begun to play an even more critical role in pushing capabilities beyond their current limits. In this work, we review photonics for AI and datacenter applications, the current landscape of multimaterial integration in photonics, and the ways in which multimaterial integration techniques have been recently utilized to push the performance of modulators on silicon and chip-scale optical frequency combs.

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## 1. Introduction

Although integrated photonics got its start in the 1960s not long after the inception of integrated electronics, it has seen much less explosive growth in the intervening decades [1]. Among a number of other factors, a key reason for this disparity is that integrated photonics, unlike integrated electronics, suffers from a lack of a single material platform that monolithically integrates all functionality in a single fabrication process with high yield on a wafer scale. For example, silicon photonics has emerged as the most feature-rich, high-volume manufacturing (HVM) compatible process but notably lacks an ideal option for integrating lasers [2,3]. It is also not the most high-performance option for critical components such as low-loss waveguides [4] or high-speed/high-efficiency modulators (discussed in section 3.1). As a result, multimaterial integration is a critical research direction for integrated photonic systems that hope to maximize performance and functionality in HVM-compatible processes. This has been underscored recently by a growing interest from industry into integration of novel materials onto silicon photonics, most notably the active development of III-V integration onto silicon by several major foundries including AIM Photonics, Imec, Intel, and Tower Semiconductor. This signals that industry leaders believe that the potential performance improvements offered by these new technologies makes the investment to overcome their many integration challenges worthwhile.

In this work we review emerging technologies that leverage multimaterial integration to push the performance and capabilities of integrated photonic technology. We choose to focus on innovations that have high impact for datacenters [5,6] and AI [7,8], where integrated photonic solutions are expected to play the greatest role in shaping the future of the modern computing landscape. The organization is as follows: The remainder of the introduction gives context for the use of photonics in AI and datacenter applications, focusing on recent trends, current challenges, and reasons to believe that multimaterial integration will be critical to the future of technology in these areas. Then in section 2 we overview multimaterial strategies utilized for integrated photonics, using III-V lasers on silicon as a means to compare and contrast. Finally, in section 3

we highlight recent developments in integrated photonics that leverage multimaterial integration to offer improvements in capability for the future of these applications.

### 1.1. *Photonics for AI acceleration*

As demand for AI computation has grown exponentially in recent years, photonics has come to the forefront as a potential platform for creating next-generation AI computing platforms that can out-perform existing and even emerging electronic systems [7,8]. Although less conventional computational structures such as spiking photonic neural networks (PNNs) [9,10] or photonic reservoir computers [11] have been explored, the majority of the attention has been placed on so-called continuous-time photonic neural networks [7] where the photonic neurons in the steady state perform the same operations as neurons in the neural networks that power all modern AI applications from image classifiers to large language models:

$$y_i(t) = \sigma \left( \sum_{j=1}^N w_{ij} x_j(t) + b_i \right) \quad (1)$$

where  $x_j(t)$  represents input signals incoming to the neuron from source  $j$  of  $N$  sources,  $w_{ij}$  represents the weight for neuron  $i$  from source  $j$ ,  $b_i$  is an optional biasing term,  $\sigma$  represents that neuron's nonlinear activation function, and  $y_i(t)$  is the output of neuron  $i$ . With each neuron performing this functionality, a full neural network can be implemented by creating a multitude of neurons in whatever number and whatever configuration of interconnections is necessary for the task (e.g., feedforward, recurrent, convolutional [7]) while accessing higher bandwidths, lower latencies, and in many cases greater efficiency. This functional form is only valid in the steady state, however, so these systems become limited in speed by the time constant of their slowest component [7]. In silicon photonics, the most popular current option due to its rich set of devices available in commercial foundry processes [2], the limiting components on the chip itself are photodetectors and modulators, both of which currently have bandwidths reaching around 50 GHz [2,12]. Photodetector bandwidth of over 250 GHz is on the horizon, with a demonstration using the same materials as are used in silicon photonics processes [13], but no such 100 GHz+ modulator is on the horizon using silicon photonics alone. As a result, researchers are turning to novel materials and multimaterial integration techniques to push modulators to ever-higher speeds while keeping silicon photonics functionality (See section 3.1).

One critical observation that has in part led to the explosion of interest in photonics for AI acceleration in particular is that the weighted summation part of the neuron is a vector-vector dot product, and that when all  $u_i$  are computed in parallel this takes the form of a matrix-vector multiplication. Photonics is naturally suited towards such linear, parallelizable operations, and opportunities to employ photonic solutions for parallelizing this linear portion date back to as early as 1985 with free-space optical demonstrations [14]. More recently, integrated photonic solutions for implementing this part of a neural network have been explored using a variety of approaches including wavelength-division-multiplexing (WDM) with microring resonators (MRRs) [15,16], coherent Mach-Zehnder-Interferometers (MZIs) [17], and photonic crossbar arrays [18,19]. MZI-based and WDM approaches are the most commonly employed, and are reviewed in [20,21].

These architectures, however, face the challenge that implementing the entire neuron requires a nonlinear element, which is difficult with photonics alone. Although all-optical nonlinearities have been explored, these require novel materials [22–24], which require fabrication process alteration, and/or specific (usually high) optical power [25–27], which introduces additional technical challenges as systems scale to large neuron counts. As a result, many PNN demonstrations have turned to optical-electrical-optical (OEO) nonlinearities [28–31] or to implementing nonlinearity fully off-chip in electronics [17,32,33].

This has led to a growing trend of photonic "AI accelerators" where the linear part is performed in photonics (a "photonic tensor core"), the nonlinearity is done in electronics, and the label of photonic neural network is dropped entirely [18,32–38]. This is done with the justification that photonic solutions are expected to out-perform electronic solutions in the limits of large processor sizes, large vector sizes, and low noise precision [39]. The goal with these systems is to specifically optimize energy consumption and throughput in order to usurp modern digital electronic solutions for AI computations. If successful, the next generation of AI datacenters will see GPUs and TPUs wholly replaced with photonic AI accelerators.

The metrics reported for these devices vary, but are mostly based on the number of operations (sometimes specifically multiply-accumulate operations or MACs) performed per second, such as Tera-Operations-Per-Second (TOP/s or TOPS), and per second per watt, such as Tera-Operations-Per-Second-Per-Watt (TOP/s/W or TOPS/W). A comparison of integrated photonic AI accelerator/photonic tensor core demonstrations is given in Table 1. Although the speed of these systems is often limited by data movement or DAC/ADC and varying precision makes direct comparison difficult [38], the main metrics of speed and/or power efficiency motivate exploration of faster and/or more efficient modulators beyond what is currently available.

**Table 1. Comparison of recent integrated photonic AI accelerator experimental demonstrations, with a representative digital processor included as an order-of-magnitude comparison point. Differing precision and I/O characteristics make direct comparison challenging, but are meant to demonstrate competitiveness with modern digital electronic processors and motivate further investigation into photonic AI acceleration.**

| Ref (year)  | Computational Architecture                | Computing Material Platform      | Light Source                | Key Performance Metric                        |
|-------------|-------------------------------------------|----------------------------------|-----------------------------|-----------------------------------------------|
| [18] (2021) | Crossbar array                            | Silicon Nitride                  | Off-chip micro-comb         | 2 TMAC/s                                      |
| [34] (2022) | WDM and Time-Delay Convolutional          | Silicon                          | Four-channel off-chip laser | 480 GOP/s                                     |
| [32] (2023) | WDM and Time-Delay Convolutional          | Silicon                          | Off-chip micro-comb         | 2.38 TOP/s/W                                  |
| [35] (2023) | WDM                                       | Silicon                          | Four off-chip lasers        | 200 kMAC/s (200 GMAC/s projected) 25 MMAC/s/W |
| [36] (2024) | MZI-based and Diffractive                 | Silicon                          | Single off-chip laser       | 160.82 TOP/s/W                                |
| [37] (2024) | MZI-based (Combination with WDM proposed) | Thin-film Lithium Niobate (TFLN) | Hybrid integrated laser     | 120 GOP/s (520 GOP/s projected)               |
| [33] (2025) | MZI-based                                 | Silicon                          | Multiple Off-chip lasers    | 2.38 TOP/s/W                                  |
| [38] (2025) | MZI-based                                 | Silicon                          | Eight off-chip lasers       | 262 TOP/s Peak 1.75 TOP/s/W Peak              |
|             | NVIDIA H100 FP16 (Peak)                   | Digital Electronics              | N/A                         | 990 TFLOP/s 1.41 TFLOP/s/W                    |

This list of demonstrations also highlights a key challenge that exists with such accelerators, which is that the inability of silicon to support lasers means they mostly rely on external lasers coupled via optical fiber [18,32–36,38]. The issue with this is that existing methods for coupling external lasers to photonic chips all require difficult active alignment, add enormous amounts of additional footprint (e.g., with V-groove arrays), or both [40]. As a result, researchers are looking to alternative methods for integrating light sources onto silicon. These efforts have been extensive, and have been extensively reviewed as well [3,41–46]. This work has recently started to come to fruition for AI applications, with [37] representing the first demonstration of a photonic tensor core with a hybrid integrated laser resulting in all optical components co-integrated at the single-chip scale rather. Additionally, progress has been made towards chip-scale "microcomb" sources for WDM-based accelerators [18,32]. Such sources which show promise for integration with WDM silicon photonic systems are reviewed in section 3.2.

### 1.2. Photonics for datacenter applications

Since the early 2000s, the exponential growth of Internet Protocol (IP) has driven data centers into the Zettabyte Era [47] recently reaching 175 ZB per year [5] with projections of 400 ZB per year in 2030 [5]. This growth is due mainly to increasing numbers users from developing nations, 80% of traffic being video of ever-increasing quality, and 50% of traffic being Machine-to-Machine (M2M) enabling connections to scale beyond total number of possible human users [48]. This increase in throughput has resulted in a proportional increase in energy consumption, resulting in over 4000 TWh per year of energy being dedicated to information and communication technology alone [5]. This increase in IP traffic has continually strained data centers, resulting in a critical need to increase throughput and energy efficiency. In the early stages, it was clear that the fundamental limits of electrical I/O and copper links were going to bottleneck the growth in IP. Therefore, in 2007 Google introduced the first incarnation of optical communication in its data centers, small form-factor pluggable (SFP) transceivers which included a 10 Gb/s vertical-cavity-surface-emitting laser (VCSEL) and multimode fiber [49]. The introduction of high-bandwidth SFPs has since opened a large market for integrated photonics, reaching an estimated 3.9 B USD in 2025 [6], and the commercial success has started a much-needed positive feedback loop between innovation and commercial deployment. Over the years transceiver technologies improved from the initial 10 Gb/s [49] to 40 Gb/s QSFP+, 100 Gb/s QSFP28, 400 Gb/s QSFP-DD, then 800 Gb/s OSFP [50], and current research has demonstrated 1+ Tb/s [51–53].

Given the goal of increasing data transmission rates, one of the most direct avenues for further development lies with modulation since increasing modulation rates directly results in more data throughput per carrier wavelength. Multimaterial integration is proving invaluable within this respect, enabling unprecedented speeds and efficiencies beyond what has been previously possible with directly-modulated lasers or silicon-based modulators. For example, recent demonstrations of complete communication links have achieved 200+ Gb/s per wavelength in silicon-compatible BaTiO<sub>3</sub> [54,55] and 400+ Gb/s in thin-film Lithium Niobate (TFLN) [52]. A summary of these demonstrations is given in Table 2 and a review of novel high-performance modulation technologies integrated onto silicon, separate from their employment in full communications links, is given in section 3.1. These modulation improvements lay a clear path towards 1.6 Tb/s SFP transceivers, but to go beyond this threshold on the chip scale requires the development of sources with many well-separated spectral lines. Frequency comb sources are ideal for this purpose, since they can generate many high-quality channels at equally-spaced wavelengths. Although frequency-comb-based systems have reached aggregate data rates of over 40 TBps [56], to date all such demonstrations (reviewed in [57]) have relied on discrete optical components. In order to support integration into SFPs in the future, frequency comb sources which are fully chip-scale will be of critical importance. These are reviewed in section 3.2.

**Table 2. Key performance metrics for photonic transceivers. Listed metrics are the best demonstrated across the material platform, and are not all achieved simultaneously.**

| Material                            | Light Source       | Per-Channel Bit Rate (Gb/s) | Energy per Bit (fJ/bit) | Transmission Distance <sup>a</sup> (km) | Optical Band(s)         |
|-------------------------------------|--------------------|-----------------------------|-------------------------|-----------------------------------------|-------------------------|
| InP VCSEL                           | Self               | 224 [58]                    | 100 [59]                | 5 [60]                                  | 850, 1060, O, C [61,62] |
| Silicon                             | External or Hybrid | 300 [63]                    | 9 [64]                  | 80 [65]                                 | O, C [66,67]            |
| III-V on Silicon                    | Heterogeneous      | 106 [68]                    | 1750 [69]               | 2 [70,71]                               | O, C [68,70,71]         |
| Thin-film LiNbO <sub>3</sub> (TFLN) | External or Hybrid | 350 [72]                    | 5.1 [73]                | 10 [72]                                 | O, C [72–74]            |
| III-V on LiNbO <sub>3</sub>         | Heterogeneous      | 600 [75]                    | 9.6 [75]                | 80 [75]                                 | O [75]                  |
| Barium Titanate BaTiO <sub>3</sub>  | External or Hybrid | 262 [76]                    | 9 [54]                  | 10 [77]                                 | O, C [54,76]            |

<sup>a</sup>Maximum transmission distance without amplification.

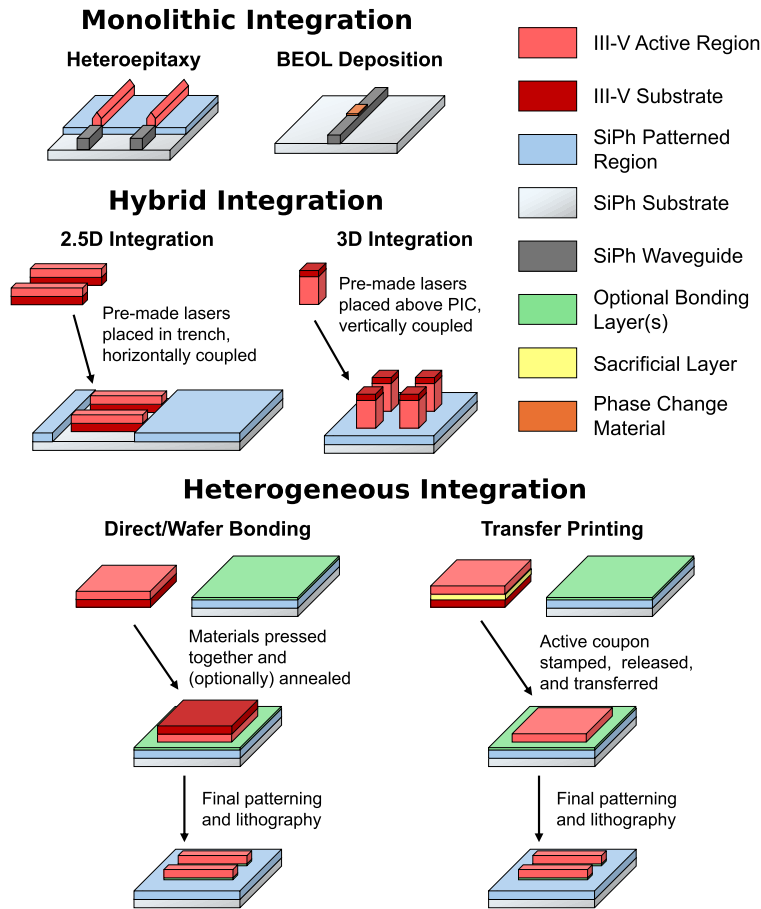
Following the ubiquitous implementation of SFPs, the next datacenter bandwidth bottleneck to naturally address was data switching. There are two approaches to optical switching, both of which have become relatively mature: free-space and integrated photonic. Free-space optical switching has been commercialized using micro-electro-mechanical systems (MEMS) [78], beam-steering [79], and liquid crystal on silicon (LCOS) [80]. Integrated photonic switches have been commercialized using microring resonators (MRR) [80] and Mach-Zehnder interferometers (MZI) [81] as the primary switching mechanism, while research-grade approaches have begun utilizing approaches leveraging plasmonic-photonics [82] and photonic crystals (PhCs) [83]. Free-space switching enables much greater channel counts compared to integrated photonic approaches, up to 347 x 347 ports [78] and up to 64 x 64 ports [81] respectively, although free-space approaches require much larger components and face cost/complexity challenges related to installation, calibration, and high voltage control [47].

## 2. Multimaterial integration strategies for photonics

Although terminology can differ from source to source, multimaterial integration strategies utilized for integrated photonics generally fall into three main categories (Fig. 1):

- **Monolithic integration:** All fabrication steps are carried out on the same substrate
- **Heterogeneous integration:** Multiple substrates with different material platforms are combined during fabrication
- **Hybrid integration:** Multiple separate photonic integrated circuits (PICs) are each fabricated separately then combined during packaging

In this section we overview each strategy in the context of integrated photonics to act as a primer on the landscape of current approaches. As examples of these integration strategies in action, we will highlight ways in which researchers have used each approach to attack integration of laser sources with silicon photonic integrated circuits (PICs). Accomplishing such integration in a high-volume manufacturing (HVM) compatible way is somewhat of a current holy grail of integrated photonics, since modern silicon photonics processes offered by foundries offer



**Fig. 1.** Visual overview of multimaterial integration strategies, focusing on examples of integration onto silicon photonics (SiPh). Drawings are not to scale.

rich features but notably cannot support light sources (lasers or optical amplifiers) directly [2]. For more thorough review of lasers integrated with silicon photonics we refer readers to other excellent recent reviews on the subject [3,41–46]. Table 3 of [3] in particular provides a direct comparison of the different techniques for the case of light sources for HVM-compatible photonics.

### 2.1. Monolithic integration

Monolithic integration is essentially the default strategy for multimaterial integration, with commercial foundries around the world offering fabrication services using processes that incorporate wide ranges of materials onto a single substrate. It is so commonplace that it is most typical to see reference to "material technologies" or "material platforms" which incorporate a whole set of individual materials that can be monolithically integrated together. Some well-known examples of mature monolithic material technologies for photonics are indium phosphide (InP) [84], silicon nitride (SiN) [4], and silicon photonics [2]. Readers are cautioned to always check the specific material stack in use, however, since these blanket terms can be non-mutually-exclusive and evolve over time. For example, silicon photonics technology began with silicon-on-insulator (SOI) as the only waveguide technology [85], but more recently incorporation of silicon nitride

waveguide layers into the same material stack has been explored by researchers [86] and even offered by commercial foundries in multi-project wafer (MPW) services [2]. For overviews of different material technologies, we refer interested readers to other excellent works which discuss near-horizon options for expanding silicon photonics [3], options for nonlinear optics [87], and options for integrated microwave photonic processing (IMWP) [88].

Unfortunately not all materials can be monolithically integrated with each other without issue. Perhaps the most well-known and progress-impeding example of such an issue is the high density of crystalline defects encountered when attempting to epitaxially grow III-V materials on silicon to create lasers or optical amplifiers. Due to mismatches in lattice constant, coefficients of thermal expansion (CTE), and polarities between silicon and III-V materials, defects arise which both act as non-radiative recombination centers and which grow with the device [41]. These can severely limit luminescence efficiency and device lifetimes, effectively eliminating any potential usefulness. To overcome such defects in monolithic integration, researchers have devised a variety of approaches including use of an "asymmetric step-graded filter" to reduce threading defect density (TDD) [89], incorporation of quantum dots to reduce both defects and sensitivity to defects [46,90], use of "aspect ratio trapping" where high aspect ratio growth is used to confine defects outside of the active region [91–93], and selective growth methods such as epitaxial lateral overgrowth (ELOG) [91,94] or template assisted selective epitaxy (TASE) [95,96]. Due to the possibility for wafer-scale processing, monolithic integration is a promising candidate for high-volume manufacturing, although monolithically integrated III-V devices still suffer from yield issues common to III-V platforms which can force high-component-count PICs to either build in redundancy or suffer poor scaling of yield with component count [3].

Beyond light source integration, post-foundry monolithic integration has become a fruitful avenue for researchers to pursue innovations leveraging new material integration with silicon as well. This allows researchers with limited in-house fabrication capabilities to leverage the rich features and process maturity offered by foundries even without being intimately connected to the foundries themselves. This is also often done with easy-to-process and widely-compatible materials, such as organics which can be added via physical vapor deposition (PVD) in an amorphous thin film. One example is the incorporation of phase-change materials (PCMs) and photochromic materials on silicon for non-volatile photonics [97–105], where deposition onto waveguides enables actuation of photonic devices which maintain their state without the need for constant power. Another is so-called silicon organic hybrid (SOH) or plasmonic optical hybrid (POH), where organic electro-optic (EO) materials enable novel high-performance modulation (discussed in section 3.1). These processes, however, are limited to those which will not cause adverse effects (e.g., annealing out prior doping by using high temperatures).

## 2.2. *Heterogeneous integration*

Heterogeneous integration in photonics refers to fabrication processes where two or more material technologies that initially existed on their own separate substrates are combined into a single PIC as an intermediate fabrication step [45,106]. This can sidestep the most catastrophic monolithic integration issues when all of the materials involved have their own separate high-quality fabrication/growth processes, such as with silicon and  $\text{LiNbO}_3$  or III-V materials [106,107], at the cost of a separate set of challenges associated with the chosen heterogeneous integration process. Heterogeneous integration can be carried out at wafer scale, making it a promising candidate for high-volume manufacturing once challenges associated with yield can be addressed [3]. There are two main approaches to heterogeneous integration, direct bonding and transfer printing:

**Direct bonding**, also referred to as wafer bonding when whole wafers are used, refers to a process by which two sufficiently flat solids are adhered to one another via contact [108]. In practical context, this means that wafer-to-wafer, die-to-wafer, or die-to-die bonding is done in

the simplest case by physically pressing the two prepared substrates against one another. The primary challenges for this approach are the need for a high degree of flatness (often sub-nm roughness [109]), necessitating pre-processing of the wafers/dies to be bonded to achieve such flatness, and the need to purge trapped air, usually necessitating high-temperature annealing or other custom outgassing processes when high temperatures cannot be tolerated. Two common approaches to ameliorate some of these strict requirements are to include an adhesive bonding layer between the two materials [110] or to include thin depositions of an interfacial material such as  $\text{Al}_2\text{O}_3$  on both sides of the bond [111]. Even under ideal conditions, however, these processes can still be vulnerable to trapped particulates which cause large areas to fail to bond [108].

**Transfer printing** is a slightly different approach where instead of a wafer or die being directly bonded to another substrate, a release layer is incorporated into one of the layer stacks so that a patterned device can be grabbed with a stamp and transferred/bonded to another material stack without taking the substrate along with it [112–114]. This is a more complex process than direct bonding/wafer bonding and similarly requires superb flatness of the receiving substrate, but carries some notable advantages. In particular, the fact that single coupons/devices are being transferred rather than large areas of material means that it is possible to choose only known-good devices or only the highest quality material to transfer, and that trapped particulates only cause one device to fail rather than a large bonded area [113]. Significant progress in transfer printing research and development has also made it possible to carry out transfer printing with micron-scale accuracy or better [115] and at wafer-scale [112].

### 2.3. Hybrid integration

Hybrid integration refers to strategies where two or more photonic chips are fabricated separately and in their entirety, then packaged together such that they can optically couple to one another [45]. Since this is a packaging-step approach it can often be simpler than heterogeneous integration, since it requires at most minor changes in the fabrication processes for the constituent devices to be integrated together. Since devices are completed prior to integration, they can often have some component of their functionality tested prior to integration as well. For example, lasers hybrid integrated with silicon photonics can be pre-tested and only known-good lasers can be chosen for integration, sidestepping issues with yield in III-V devices. Hybrid integration takes many forms but can largely be separated into two families of approaches, 2.5D and 3D integration. We choose to adopt this terminology from [3].

**2.5D integration** refers to integration where two devices are placed side-by-side and optically coupled, usually with one die in a trench on the other die or both dies on a common sub-mount. The nomenclature "2.5D" comes from the fact that the vertical registration is generally set by the fabrication process, then active alignment or post-processing is carried out to precisely place the incoming device in its place in 2D. Many approaches for this type of integration have emerged for III-V laser integration specifically, including butt-coupling [116,117], contact coupling [118], photonic wire bonding [119–121] or other packaging-step coupler waveguide deposition [122]. High-volume processes leveraging 2.5D integration have emerged as well [123]. The primary difficulty of such approaches lies in optical coupling, which necessitates monitored alignment to achieve sufficient precision [118], post-processing [121,122], or additional components [123].

**3D integration** refers to integration where devices are placed vertically on top of one another and made to optically couple. In the case of III-V lasers 3D hybrid integrated with silicon photonics, this usually takes the form of surface-emitting lasers with light vertically coupled to silicon waveguides via grating couplers. Examples include flip-chip bonded HCSELs [124], flip-chip bonded VCSELs [125–128], and transfer printed VCSELs [129]. A distinguishing feature of 3D integration is that in using a combination of large-alignment-tolerance grating couplers and physical structures for mechanical registration, alignment can be done without the

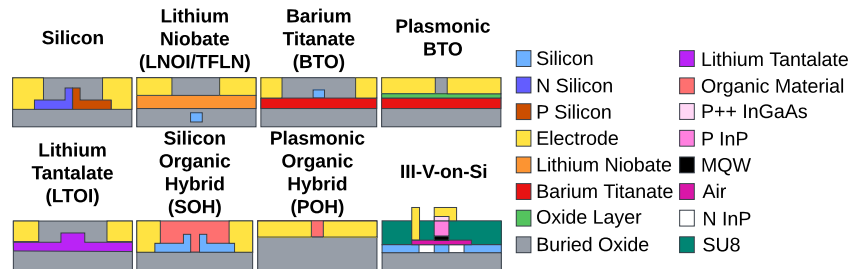
need for active monitoring or post-processing [126,128] although registration structures add complexity and grating couplers are generally more lossy than other options.

### 3. Emerging technologies enabled by multimaterial integration

In this section we overview recently developed technologies with high impact for datacenter and AI applications that have been made possible through the use of the multimaterial integration techniques we overviewed in the previous section. Although the topic of light sources on silicon photonics platforms is of critical importance to the future of datacenters and AI, this has been reviewed extensively already so we refer interested readers to existing recent reviews for more information on the topic [3,41–46].

#### 3.1. High-speed and high-efficiency modulators on silicon

Optical communication links and interconnections are critical components for datacenters and AI compute systems, and demand continuously increasing transmission speed and efficiency to keep up with continuously growing demand for computation. At the heart of these systems are modulators, which are responsible for encoding data onto optical carriers. To support this continued growth, there is correspondingly great interest in improving the efficiency and speed while maintaining the feature set offered by silicon photonics. Multimaterial integration in modulators is a promising direction, since materials outside of current foundry-standard processes can offer improvements in these areas. Of particular interest are ferroelectric phase modulators (e.g., in  $\text{LiNbO}_3$ ), plasmonic and silicon organic hybrids (POH and SOH), and III-V-on-silicon using the plasma dispersion effect (Fig. 2, Table 3). The primary metrics of interest are modulator bandwidth, listed herein as 3 dB bandwidth  $f_{3\text{dB}}$ , and efficiency in the form of voltage-length product  $V_\pi L$  where  $V_\pi$  is the full-on-full-off voltage swing and  $L$  is the device length.  $V_\pi L$  can be a useful point of comparison for footprint in the case of common modulator structures such as MZIs as well since active region length dominates the footprint in these cases.



**Fig. 2.** Cross sections for state-of-the-art modulators demonstrated for integration with silicon photonics. Drawings are not to scale.

**P-N doped silicon** is the material most readily available for high-speed modulators in commercial foundries today [2]. This technology and its various architectures have been extensively studied, with some schemes demonstrating bandwidth as high as 50 GHz and  $V_\pi L$  as low as 0.52 V·cm [12,141]. Silicon modulators are based on the plasma dispersion effect, however, which fundamentally limits bandwidth due to free carrier dynamics and results in unwanted insertion loss. Furthermore, while efficiency can be improved by sacrificing bandwidth and loss, there has not yet been a favorable demonstration that improves all critical metrics simultaneously [142–144]. In order to address these fundamental challenges in silicon-based modulators, several options utilizing integration of novel materials onto silicon have been explored to improve state-of-the-art performance.

**Table 3. State of the art demonstrated modulator bandwidth and efficiency. Additional advantages and disadvantages are mentioned throughout section 3.1 where applicable.**

| Material Platform          | Bandwidth (GHz) | $V_{\pi}L$ (V·cm) |
|----------------------------|-----------------|-------------------|
| Silicon                    | 50 [12]         | 0.52 [12]         |
| LNOI/TFLN                  | 110 [130]       | 2.2 [131]         |
| BTO                        | 30 [111]        | 0.45 [111]        |
| Plasmonic-BTO <sup>a</sup> | 270 [132]       | 0.0045 [133]      |
| LTOI                       | 110 [134]       | 2.88 [134]        |
| SOH                        | 100 [135–137]   | 0.0084 [136]      |
| POH <sup>a</sup>           | >110 [138]      | ~0.005 [139]      |
| III-V-on-Si                | 40 [140]        | 0.005 [140]       |

<sup>a</sup>Plasmonics come with the cost of worse insertion loss and more difficult fabrication.

Ferroelectric materials such as lithium niobate ( $\text{LiNbO}_3$ ), barium titanate ( $\text{BaTiO}_3$  or BTO), and lithium tantalate ( $\text{LiTaO}_3$ ) have garnered attention for their exceptionally high electro-optic (EO) coefficients which allow for high-speed and high-efficiency modulators. Heterogeneous integration onto silicon has also been demonstrated for each platform, allowing for flexibility in process where the ferroelectric may be patterned or unpatterned and optical modes may exist only in the ferroelectric during modulation or exist as a hybrid silicon/ferroelectric mode. They have become increasingly commercially available both as individual platforms and integrated onto silicon as well, although issues with CMOS foundry compatibility pose serious challenges for the prospect of HVM [145–147]. Nonetheless, recent years have seen heavy investment into ferroelectric-on-insulator process development, signaling that the need for improved performance is enough to drive the investment required to overcome manufacturing challenges.

**Lithium Niobate ( $\text{LiNbO}_3$ )** is the most thoroughly-explored of the ferroelectrics, with initial exploration dating back to as early as 1974 [148]. More recently, this has evolved into  $\text{LiNbO}_3$  on insulator (LNOI), also referred to as thin-film lithium niobate (TFLN), which has thin film  $\text{LiNbO}_3$  heterogeneously integrated onto a silica buried layer which sits on a substrate [107]. This has enabled the fabrication of ridge structures on LNOI platforms which has in-turn enabled high index contrast and flexibility of fabrication [149]. This integration strategy can result in a variety of material stacks including silicon and silicon nitride waveguides [130,131,150–152]. [131] reported the first wafer bonded LNOI platform that fully transitions the optical mode from silicon waveguide to an etched  $\text{LiNbO}_3$  waveguide through vertical adiabatic couplers, demonstrating record  $V_{\pi}L$  of 2.2 V·cm. Record bandwidth of 110 GHz was achieved in [130] with a non-patterned  $\text{LiNbO}_3$  layer on patterned silicon that utilized a hybrid mode extending into both materials. Such a stack is a particularly attractive option for scaling LNOI/TFLN towards HVM.

**Barium titanate (BTO)** is a relatively new ferroelectric that is being explored for enabling higher modulation speeds with an electro-optic coefficient even larger than  $\text{LiNbO}_3$  (over an order of magnitude larger for bulk material comparison) [111,153,154]. BTO integrated with silicon photonics has been demonstrated with monolithic integration [155–157] and heterogeneous integration through wafer bonding onto silicon [111,147,158]. Although BTO performance has been modest so far, with [111] demonstrating 30 GHz bandwidth and a  $V_{\pi}L$  of 0.45 V·cm, the possibility of monolithic integration and projected bandwidth of over 260 GHz [159] has made BTO an attractive platform for continued investigation.

**Plasmonic-BTO** platforms have also been investigated experimentally, aiming to maximize compactness and efficiency over BTO alone at the cost of greater insertion loss and requiring more difficult fabrication. [160] presents a monolithically integrated, non-patterned modulator with a  $V_{\pi}L$  of 0.1 V·cm with a bandwidth exceeding 150 GHz. This is done with the combination of a high electro-optic coefficient enabled by BTO and a small footprint enabled by the plasmonic modulator. [133] integrated BTO into a standard SiN foundry process, reporting a heterogeneously integrated, patterned modulator with a flat frequency response beyond 70 GHz and a  $V_{\pi}L$  of 0.0045 V·cm. Furthermore, a plasmonic, monolithically integrated, nonpatterned phase shifter has been experimentally shown to have a consistent electro-optic response up to 270 GHz [132].

**Lithium Tantalate (LiTaO<sub>3</sub>)** has emerged as an even more recent candidate for high speed modulators owing to optical properties such as an EO coefficient nearly identical to LiNbO<sub>3</sub> with a higher optical damage threshold and a smaller optical birefringence [161]. [134] has demonstrated a LiTaO<sub>3</sub> on insulator (LTOI) modulator with 110 GHz of bandwidth and a  $V_{\pi}L$  of 2.88 V·cm showing promise for this platform.

**Silicon Organic Hybrid (SOH)** modulators have become an active area of research. These modulators consist of organic EO materials deposited onto waveguides, and exhibit favorable EO characteristics while ensuring ease of integration through simple back-end-of-line (BEOL) compatible deposition techniques. These modulators typically feature a silicon slot waveguide with the material deposited in the slot for maximum mode overlap with the organic material. Recent work in SOH has demonstrated >100 GHz bandwidth [135–137] and modulator efficiencies as low as ~0.0084 V·cm [136].

**Plasmonic organic hybrid (POH)** platforms also utilize strong EO effects from organics, but use plasmonic effects to achieve better efficiencies and much smaller footprints at the cost of greater insertion loss and more difficult fabrication. These modulators typically feature a taper from an optical waveguide to a plasmonic slot which allows for subwavelength confinement [162]. One of the first POH modulators demonstrated a bandwidth of at least 65 GHz and achieved a  $V_{\pi}L$  of ~0.2 V·cm [163]. Further engineering has resulted in demonstrations of bandwidths above 110 GHz [138] and  $V_{\pi}L$  as low as ~0.005 V·cm [139] in POH.

For both organic platforms, the main concern is long term poling stability. Organic polymers with strong EO coefficients such as JRD1 used in the cited work have an intrinsically low glass transition temperature of around 80 degrees C, which is below the Telcordia standard of 85 degrees C [164]. High temperatures relative to the glass transition temperature lead to depoling of the dipoles in the material, causing modulators to lose effectiveness over time. In order to combat this issue, recent work on SOH platforms has used techniques such as host matrix embedding (amorphous polycarbonate) [165], backbone side chains (methyl methacrylate polymer) [166], and cross-linking organic polymers (HLD1 and HLD2) [167,168]. Glass transition temperatures as high as 194 degrees C have been demonstrated while retaining modest EO coefficients [169].

**III-V materials on silicon** are heavily sought after for their potential as on chip light sources, but they have also found a potential application enabling more efficient modulators on silicon via III-V/Si MOS capacitor phase shifters. Much like a doped silicon modulator, these modulators utilize the plasma dispersion effect. However, by using a III-V material which has an order of magnitude stronger plasma dispersion effect, a more efficient modulator is possible with less stray absorption [170,171]. Record performance was demonstrated using an electroabsorption modulator (EAM) based on an InP/multiple quantum well (MQW) device that was micro-transfer printed onto silicon photonic chip, and achieved bandwidth up to 40 GHz with a  $V_{\pi}L$  as low as 0.005 V·cm [140].

Several other multi-material modulation technologies have been proposed, such as silicon germanium electro-absorption modulators [172–174] and 2-D material on silicon modulators [175,176] although these are currently not competitive for datacenter and AI applications. Non-volatile material approaches, which only require power when switching, such as liquid crystals

[177,178], phase-change materials (PCMs) [102,179], and nano-opto-electro-mechanical systems (NOEMs) [180,181] have also been explored, but are more useful for memory and low-bandwidth (e.g., switching) applications.

### 3.2. Fully-integrated comb sources

Wavelength-division-multiplexing (WDM) is a core technique for scaling interconnections in both datacenters [5] and certain varieties of photonic AI accelerators [15], where multiple signals can be encoded onto multiple wavelengths of light and sent through a single waveguide without interacting. From a light generation perspective, the main challenge of implementing a WDM system lies in delivering multiple wavelengths of light to a photonic chip. The ideal WDM optical spectrum is a series of narrow lines, one for each channel (e.g., the ITU grid for DWDM in the optical C-band). An obvious solution is to use an individual laser for each channel, but as channel count scales up this becomes impractical even with on-chip lasers. Instead, researchers are increasingly turning towards optical frequency combs. These are sources whose spectra consists of a series of equidistant lines, sometimes thousands at once [182,183], and outside of datacenters and AI they have previously been leveraged to revolutionize timekeeping, spectroscopy, metrology, and more [87,183,184].

Of particular interest are fully-integrated comb sources that emit in the optical O-band or C-band, since these will be natively compatible with existing silicon photonics infrastructure and will therefore be enablers for new WDM-based applications [87,183]. Such comb sources generally fall into two categories: integrated mode-locked lasers (MLLs) and nonlinear Kerr frequency combs. Table 4 gives a comparison of recent experimental demonstrations of fully-integrated chip-scale O/C-band comb sources and their core spectral properties, focusing on designs featuring tunable parameters.

**Table 4. Optical O/C-band fully-integrated frequency comb demonstrations and their key metrics, only including demonstrations where all components are co-integrated. Bandwidth values for Kerr combs exclude the pump wavelength. Monolithically-integrated MLLs are not listed due to their lack of tunability, a critical weakness for the prospect of use in WDM systems. We instead refer readers to [192,193] for reviews including monolithic MLLs.**

| Category             | Ref (year)   | Integration Type & Material Platforms                | Line Spacing (GHz) | Bandwidth (metric)                              | Pump/center wavelength (nm) |
|----------------------|--------------|------------------------------------------------------|--------------------|-------------------------------------------------|-----------------------------|
| MLL                  | [185] (2024) | Heterogeneous III-V/Si <sub>3</sub> N <sub>4</sub>   | 3                  | 23 nm (10 dB)                                   | ~1555                       |
| Kerr (Anomalous GVD) | [186] (2020) | Hybrid III-V/Si <sub>3</sub> N <sub>4</sub>          | 40                 | 20 nm <sup>a</sup> (10dB sech <sup>2</sup> fit) | 1556                        |
|                      | [187] (2021) | Heterogeneous InP-/Si/Si <sub>3</sub> N <sub>4</sub> | 100                | 20 nm <sup>a</sup> (10dB sech <sup>2</sup> fit) | ~1550                       |
|                      | [188] (2024) | Hybrid III-V/Si <sub>3</sub> N <sub>4</sub>          | 300                | 1.44 THz (FWHM sech <sup>2</sup> fit)           | ~1557                       |
| Kerr (Normal GVD)    | [189] (2021) | Hybrid III-V/Si <sub>3</sub> N <sub>4</sub>          | 43.2               | 3.5 nm <sup>a</sup> (10 dB)                     | 1555.4                      |
|                      | [190] (2022) | Hybrid III-V/Si <sub>3</sub> N <sub>4</sub>          | 26.2               | 1.0 nm <sup>a</sup> (10 dB)                     | 1556                        |
|                      | [191] (2025) | Hybrid III-V/Si <sub>3</sub> N <sub>4</sub>          | 10.7               | 0.75 nm <sup>a</sup> (10 dB)                    | 1550                        |

<sup>a</sup>Values estimated from reference figures.

Modelocked lasers (MLLs) are lasers that output coherent combs by synchronizing the phases of multiple longitudinal modes of a resonant cavity [194]. Integrated solutions typically accomplish this by using a saturable absorber within the gain region, implemented by reverse biasing a section of the laser gain material. Monolithically-integrated MLLs in the optical C band around 1.55  $\mu\text{m}$  center wavelength were first demonstrated in 1989 [195], and continued to improve

over several decades [192], but have a fundamental flaw for use in WDM systems which is that the laser cavity mirrors come from cleaved facets. The comb line spacing (free-spectral range/FSR) is set, in part, by the intra-cavity distance, meaning cleaved-facet modelocked lasers do not have an easy mechanism by which to tune line spacing post-fabrication. This has caused researchers to consider heterogeneous integration with material platforms supporting low-loss passive waveguides, where tunable-effective-length cavities can be implemented. These were first demonstrated with III-V-on-Si [192,196,197], although none of these were in the optical C or O bands. Later, researchers turned to III-V-on-SiN to access even lower losses [185,198,199]. [185] is a particularly important result for the prospect of use in WDM systems since it is the first demonstration operating in the C band. If optimized, this is expected to support over 1000 comb lines over a wide band (23 nm). MLLs generally provide the lowest line spacing, with as low as 755 MHz demonstrated just outside of the C band [199]. This can be desirable for certain applications, but is not always useful for WDM communications or computing since even at 100 GHz spacing this can require de-interleaving to be compatible with other components [200].

Kerr frequency combs instead leverage nonlinear optical processes, primarily four-wave mixing (FWM) and the creation/synchronization of solitons [183]. In order to implement such a process on an integrated photonic platform, a resonator is created in a material with a sufficiently high combination of nonlinear Kerr coefficient and readily-available ultra-high Q factor resonators [184]. This makes Silicon Nitride a particularly attractive option, although it does not support lasers and therefore must be hybrid or heterogeneously integrated in order to form a chip-scale system. All fully-integrated chip-scale C-band demonstrations to date, in fact, have leveraged III-V DFB lasers either hybrid [186,188–191] or heterogeneously [187] integrated with Si<sub>3</sub>N<sub>4</sub> (although progress has also been made towards similar demonstrations in AlGaAs-on-insulator [201]). [187] is of particular note because it heterogeneously integrates both Si<sub>3</sub>N<sub>4</sub> and Si, paving the way for a true system-on-chip leveraging Kerr combs for WDM.

Kerr combs notably come in two distinct varieties. The conventional approach is to mode lock solitons in a resonator, as in [186–188], which is referred to as a dispersive Kerr soliton (DKS) or “bright pulse” soliton approach. These can be advantageous for WDM systems since the free-spectral range (FSR) of the resonator drives the spacing, and this FSR can easily land in a convenient range for WDM of 10s to 100s of GHz. Their lines also ideally follow a  $\text{sech}^2$  falloff in power from the pump, which can lead to considerably wider combs compared to MLLs if low powers can be tolerated. DKS approaches, however, require anomalous group velocity dispersion (GVD) in the waveguide. Since the ultra-low-loss LPCVD Si<sub>3</sub>N<sub>4</sub> that one would prefer to use for ultra-high Q microresonators displays normal dispersion in the optical C-band, dispersion engineering is required to form DKS-based combs [190]. Although a number of approaches have been explored for doing this and successful demonstrations have resulted [186–188], dispersion engineering typically requires some deviation from a foundry-standard Si<sub>3</sub>N<sub>4</sub> fabrication process [202] and these are therefore of questionable suitability for future high-volume manufacturing scenarios. To utilize foundry-standard processes where GVD is normal or to simply avoid dispersion engineering, some researchers have opted to instead excite combs using “dark pulse” solitons [189,191] or “platicons” [190]. These can offer higher pump-to-comb conversion efficiency, but can be more difficult to access. They also offer a different spectral envelope, with only a few lines around the pump being bright but much larger flat “wings” with many lines at powers 20-40 dB lower [189,191]. Both the anomalous-GVD and normal-GVD approaches can also benefit from self-injection locking, where back-scatter from the microresonator is used to lock the resonator and laser [186–191]. This ameliorates strict frequency tuning requirements, and has even resulted in a demonstration of “turnkey” operation in a fully-integrated microcomb source, where the comb is available immediately upon turn-on with no additional input [186].

#### 4. Conclusion

Photonic solutions have the potential to shape the future of the modern computing landscape by continuing to push forward the capabilities and performance of datacenters and AI accelerators. In order to continue to improve performance beyond current limits and push systems towards fully chip-scale solutions, it is clear that novel integration strategies and novel material platforms are necessary. In this work we reviewed the current landscape of multimaterial integration strategies in photonics, covering monolithic, heterogeneous, and hybrid integration. We then reviewed important ways in which researchers have recently leveraged these techniques to push forward critical components of photonic systems in datacenter and AI applications. First, we discussed how novel material integration onto silicon can facilitate higher-speed and higher-efficiency modulation without sacrificing the features of silicon photonics. Then, we overviewed how integration of active materials with  $\text{Si}_3\text{N}_4$  has enabled chip-scale frequency comb sources with promise for future WDM applications. With this wealth of recent demonstrated success, mostly in the early R&D stages, we expect to see these integration techniques and technologies make their way into commercial foundry processes to revolutionize both the photonics technologies available at scale and the applications they enable.

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**Data availability.** No data were generated or analyzed in the presented research.

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